BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade) — ANSWER KEY (Version 1)

Syllabus: Unit 7 — Density and Temperature

Total Marks: 65	**Prepared for Marking Use**	**Version: 1**

SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

#	Question	Correct Answer	Explanation
1	Density is defined as:	**B**	**Mass per unit volume** ($\rho = m/V$). Option A reverses the definition. Options C and D describe pressure and gravitational field strength respectively, not density.
2	The SI unit of density is:	**C**	**kg/m^3** is the SI unit. g/cm^3 and g/mL are common but non-SI. kg/m^2 is the unit of surface density, not volumetric density.
3	A cube of side 5 cm has mass 500 g. Its density is:	**B**	**4 g/cm^3**. Volume = $(5\text{ cm})^3 = 125\text{ cm}^3$. $\rho = 500/125 = 4\text{ g/cm}^3$. Option A (2) uses half the volume by mistake. Option D (10) divides incorrectly by 50.
4	Osmium density is:	**B**	**22,590 kg/m^3** (written as 22.59 g/cm^3 in the textbook). Mercury is 13,600, gold is 19,300, and iron is 7,900. Osmium is the densest material at room temperature.
5	Volume of object in displacement method equals:	**C**	**Final volume minus initial volume** ($V_f - V_i$). The displaced water volume equals the object's volume. Option D reverses the subtraction, giving a negative result.
6	Plasma consists of:	**B**	**Positive ions and free electrons**. When a gas is heated to very high temperature, electrons are stripped from atoms, producing ions and free electrons. This ionic state is plasma, distinct from neutral gas.
7	Temperature is directly proportional to average:	**B**	**Kinetic energy of its particles**. The textbook states: "temperature of a substance is directly proportional to the average K.E. of its particles." Potential energy depends on intermolecular spacing, not temperature alone.
8	Absolute zero on Celsius scale is:	**C**	**−273 °C**. The textbook states absolute zero = 0 K = −273.15 °C. Common error: students confuse absolute zero with the freezing point of water (0 °C) or think it is +273 °C.
9	Boiling point of water on Kelvin scale is:	**C**	**373 K**. T(K) = T(°C) + 273 = 100 + 273 = 373 K. Option B (273 K) is the freezing point of water. Option D (212) is the Fahrenheit boiling point, not Kelvin.
10	37 °C converted to Fahrenheit is:	**A**	**98.6 °F**. F = (9/5 × C) + 32 = (9/5 × 37) + 32 = 66.6 + 32 = 98.6 °F. Option B (66.6) is the result before adding 32 — a classic off-by-32 error.
11	Thermometric property of liquid-in-glass thermometer:	**C**	**Volume expansion of the liquid**. Mercury or alcohol expands linearly when heated. Options A and B

			describe resistance thermometers and thermocouples respectively; Option D is liquid crystal thermometers.
12	Narrower capillary tube results in:	**B**	**Greater sensitivity and faster response**. Less liquid in the tube reacts more quickly to heat. Option A is backwards. Sensitivity is analogous to least count; a narrow bore means smaller change in temperature can be detected.

SECTION B — Short Answer Key (11 × 3 = 33 Marks)

2(i) — Density definition, formula, and scalar/vector classification **3 marks**

Definition: Density is the mass per unit volume of a substance. **1**
Formula: $\rho = m/V$. SI unit: kg/m^3 (smaller unit: g/cm^3 or g/mL). **1**
Scalar quantity: density has magnitude only and no direction; it does not require a direction to be fully described. **1**

OR

Conversion: 1 g/cm^3 = 1000 kg/m^3 (multiply by 1000). **1**
So 0.9 g/cm^3 of petrol = **900 kg/m^3**. **1**
Density of liquids is usually measured in g/mL because 1 mL = 1 cm^3, making measurement with a measuring cylinder direct and convenient without unit conversion. **1**

2(ii) — Density calculations (wooden block and metal ball) **3 marks**

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Wooden block: rho = m/V = 300 g / 150 cm³ = 2 g/cm³ [1] Metal ball: rho = m/V = 500 g / 50 cm³ = 10 g/cm³ [1] The metal ball is denser (10 g/cm³ > 2 g/cm³) [1]
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OR

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Cooking oil density: rho = 450 g / 500 mL = 0.9 g/mL [1] Volume of 1.8 kg = 1800 g: V = m/rho = 1800 g / 0.9 g/mL = 2000 mL = 2 L [1+1]
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2(iii) — Displacement method procedure **3 marks**

Steps: (1) Find mass of the irregular solid using a balance. (2) Add water to a measuring cylinder; record **initial volume V_i**. (3) Tie the solid with a thread and lower it fully into the cylinder. (4) Record the **final volume V_f**. (5) Volume of solid = $V_f - V_i$. (6) Density = mass / $(V_f - V_i)$. **2**
Formula: $\rho = m / (V_f - V_i)$. **1**

OR

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Given: mass = 80.52 g, Vi = 18 mL, Vf = 46 mL Volume of rock = Vf - Vi = 46 - 18 = 28 mL = 28 cm³ [1] Density = 80.52 g / 28 cm³ = 2.88 g/cm³ [1+1]
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2(iv) — Comparison of particle structure of solids, liquids, gases **3 marks**

Property	Solid	Liquid	Gas
Particle spacing	Very close (fixed lattice)	Close but random	Far apart
Particle motion	Vibrate about fixed positions	Move around each other	Move quickly in all directions
Density	High	Medium	Low

3 (1 per row fully correct)

OR

Liquids take the shape of their container because the **attractive force between liquid particles** is intermediate: strong enough to keep particles close together (fixed volume) but not strong enough to lock them in fixed positions. They can therefore flow and slide past each other, adopting the container's shape. **2** Solids have a very strong attractive force that holds particles in a fixed lattice, preventing flow. **1**

2(v) — Plasma: definition, formation, examples 3 marks

Definition: Plasma is the fourth state of matter consisting of **positive ions and free electrons** (an ionized state). 1

Formation: When a gas is heated to very high temperature or subjected to high pressure, electrons are stripped from atoms. The substance can also be transformed into plasma using a high electric and magnetic field. 1

Two examples from daily life: neon plasma tube lights; welding arc plasma (also acceptable: plasma lamps, lightning in the sky). 1

OR

Plasma is called the fourth state of matter because it is a distinct, highly ionized state that is neither solid, liquid, nor gas. 1

Two properties distinguishing plasma from gas: (1) Plasma can **highly conduct electric current** because it has free electrons and moving ions; gas is an insulator. (2) Plasma is formed only at **very high temperature or high pressure**, while gas exists at ordinary conditions. 2

2(vi) — Particle motion and temperature; cooling effect 3 marks

Relationship: Temperature is the measure of the hotness or coldness of a substance and is **directly proportional to the average kinetic energy of its particles**. When a substance is heated, particle speed and K.E. increase; temperature rises. 2

When cooled, the speed and kinetic energy of particles become slower and smaller, so the temperature of the substance decreases. 1

OR

Absolute zero is the lowest possible temperature of a substance at which its particles have least (zero) kinetic energy. 1 Its value on the Celsius scale is **−273.15 °C** (approximately −273 °C), which equals **0 K**. 1 At absolute zero, particles cannot collide with each other or the container walls, so pressure = 0 Pa. Since there is no kinetic energy left to remove, no further cooling is possible. 1

2(vii) — Internal energy, components, and effect of heating 3 marks

Internal energy of a substance is the total energy possessed by its particles, consisting of the total **kinetic energy** and **potential energy** of its particles. 1

Kinetic energy arises from particle motion; potential energy arises from attractive forces between particles. 1

When a substance is heated, the speed of particles increases. This increases their kinetic energy, and hence the **internal energy of the substance also increases**. 1

OR

Three forms of kinetic energy: **translational K.E.**, **rotational K.E.**, and **vibrational K.E.** 2

In an **ideal gas**, internal energy is entirely due to translational kinetic energy of particles (no rotational or vibrational contributions, no intermolecular potential energy). 1

2(viii) — Three temperature scales with fixed points and divisions 3 marks

Scale	Lower fixed point	Upper fixed point	Divisions
Celsius	0 °C (ice point)	100 °C (steam point)	100
Fahrenheit	32 °F	212 °F	180
Kelvin	273 K	373 K	100

3 (1 per complete row)

OR

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(a) 25°C to Kelvin: T(K) = 25 + 273 = 298 K [1] (b) 350 K to Celsius: T(°C) = 350 - 273 = 77°C [1] (c) 100°C to °F: F = (9/5 × 100) + 32 = 180 + 32 = 212°F [1]
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2(ix) — Sensitivity of thermometer and effect of capillary tube diameter 3 marks

Sensitivity is the ability of a thermometer to detect the smallest change in temperature; it is the smallest temperature variation it can measure. 1
A **narrower capillary tube** holds less liquid, so even a tiny amount of heat causes the liquid column to rise more noticeably. Therefore a narrower bore gives **greater sensitivity**. 1
A wider bore holds more liquid; a given temperature change produces less visible rise in the column, reducing sensitivity. Sensitivity is analogous to the least count of a measuring instrument. 1

OR

Range is the span of temperatures a thermometer can measure, defined by its lowest and highest measurable temperatures. 1
Clinical mercury thermometer range: 35°C to 42°C.

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Lower limit: F = (9/5 × 35) + 32 = 63 + 32 = 95°F Upper limit: F = (9/5 × 42) + 32 = 75.6 + 32 = 107.6°F Range in Fahrenheit: 95°F to 107.6°F [2]
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2(x) — Thermocouple thermometer structure and working 3 marks

Structure: A thermocouple thermometer has **two wires** of different metals joined at one end (the **hot junction**). The other ends of the wires are the **cold junctions**, connected to a voltmeter. 1
Working: When the hot junction is heated, it causes free electrons in the metals to flow across the junction, creating a **voltage difference** between the hot and cold junctions. 1
This voltage is linearly proportional to the temperature difference. The voltmeter is calibrated so the voltage reading corresponds to a specific temperature. 1

OR

Liquid crystal strip thermometers contain liquid crystal materials packed in a plastic strip that change colour with a change in temperature. 1
When the strip touches a body, the liquid crystal material changes colour; matching the colour to a temperature scale gives the reading. 1
Two applications: (1) fever thermometers for babies (placed on forehead); (2) monitoring temperature of aquariums and baby bottles. 1

2(xi) — Thermometric property and its types 3 marks

A **thermometric property** is a physical property of a material that varies predictably and measurably with temperature and can therefore be used to measure temperature. 1
(a) Liquid-in-glass thermometer: **volume expansion of the liquid** (mercury or alcohol).
(b) Constant-volume gas thermometer: **pressure change of a fixed mass of gas**.
(c) Resistance thermometer (thermistor): **change in electrical resistance**. 2 (1 per correct pair)

OR

Effect of bulb size on range: A larger bulb contains more liquid, giving more liquid available to expand, so the thermometer can cover a wider temperature range. 1
However, a larger bulb also responds more slowly to temperature changes.
Effect of nature of liquid on sensitivity: Mercury and alcohol expand more than other liquids (e.g., water) when heated. Thermometers using mercury or alcohol therefore show a larger rise in column per degree, making them **more sensitive**. Mercury also has a wide boiling point range, adding to its usefulness. 2

SECTION C — Long Answer Key (4 × 5 = 20 Marks)

3(i) — Density definition, particle explanation, brick calculation 5 marks

Density is mass per unit volume: $\rho = m/V$. SI unit: kg/m^3. 1

Why solids > liquids > gases in density: Density depends on how closely packed particles are. Solids have atoms in a tight fixed lattice, so many atoms fit in a small volume (high density). In liquids, atoms are close but randomly arranged with slightly more spacing (medium density). In gases, atoms are far apart with negligible attractive forces (low density). 2

Brick calculation:

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Given: m = 3.30 kg L = 21.6 cm = 0.216 m W = 10.2 cm = 0.102 m H = 6.35 cm = 0.0635 m Volume = L × W × H = 0.216 × 0.102 × 0.0635 = 0.001399 m³ ≈ 0.00140 m³ [1] Density = m/V = 3.30 kg / 0.00140 m³ = 2357 kg/m³ ≈ 2360 kg/m³ [1]
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OR

Displacement method (full detail): 3

Step 1: Find mass of stone using a balance.

Step 2: Add water to measuring cylinder; record initial volume V_i.

Step 3: Tie stone with a thread; lower it fully into the water.

Step 4: Record the new (final) volume V_f. Water has risen because stone displaced it.

Step 5: Volume of stone = $V_f - V_i$.

Step 6: Density = mass of stone / $(V_f - V_i)$.

Formula: $\rho = m / (V_f - V_i)$. 1

Calculation:

```
Given: m = 200 g, Vi = 40 mL, Vf = 73 mL Volume = 73 - 40 = 33 mL = 33 cm³ [1] Density = 200 g / 33 cm³ = 6.06 g/cm³ ≈ 6.1 g/mL [1]
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3(ii) — Four states of matter with comparison table and plasma examples 5 marks

Table comparing the three common states: 2

Property	Solid	Liquid	Gas
Density	High	Medium	Low
Particle arrangement	Fixed positions (lattice)	Random positions	Random positions
Particle movement	Vibrate about fixed position	Move around each other	Move quickly in all directions
Energy of particles	Low	Greater	Highest

Plasma formation: When a gas is heated to very high temperature (or subjected to high pressure, or a high electric/magnetic field), the kinetic energy of molecules increases so much that electrons are stripped from atoms. Atoms become **positive ions** and the freed electrons move independently. This ionic state is plasma. 2

Three examples: (1) The Sun and stars glow because of plasma. (2) Neon plasma tube lights in signs. (3) Aurora (polar lights) at the north and south poles are caused by plasma. (Also acceptable: lightning, welding arc plasma, plasma lamps.) 1

OR

Temperature and particle motion: Temperature is directly proportional to the average K.E. of particles. Heating increases particle speed and K.E. 1

Internal energy = total K.E. + total potential energy of particles. When temperature rises, internal energy increases. 1

Absolute zero: When heat is continuously removed, particle K.E. decreases. The point at which K.E. reaches zero is **absolute zero**. Value: 0 K = −273.15 °C. 1

Kinetic energy vs. Kelvin temperature graph: A straight line through the origin on a K.E. (y-axis) vs. Kelvin temperature (x-axis) graph. The line starts at the origin (0 K, 0 K.E.) and rises linearly. Absolute zero is the point where the line meets the x-axis (K.E. = 0, T = 0 K). 2

3(iii) — Temperature scale conversions (full set of calculations) 5 marks

Conversion formulas: 1

T(K) = T(°C) + 273 | T(°F) = (9/5) × T(°C) + 32 | T(°C) = (T(°F) − 32) × 5/9

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(a) Convert 37°C: Kelvin: T = 37 + 273 = 310 K [1] Fahrenheit: T = (9/5 × 37) + 32 = 66.6 + 32 = 98.6°F [1] (b) Convert 212°F: Celsius: T = (212 - 32) × 5/9 = 180 × 5/9 = 100°C [1] Kelvin: T = 100 + 273 = 373 K (included above) (c) Convert 300 K: Celsius: T = 300 - 273 = 27°C [1] Fahrenheit: T = (9/5 × 27) + 32 = 48.6 + 32 = 80.6°F (1 mark shared above)
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Mark allocation: 1 (formulas) + 1 (37°C to K) + 1 (37°C to °F) + 1 (212°F to °C) + 1 (300 K to °C or °F).

OR

Calibration of a thermometer: Calibration means making a proper, accurate scale on a thermometer by using two **fixed (reference) points**. 1

Lower reference point (ice point): melting point of pure ice = 0°C = 32°F = 273 K. **Upper reference point** (steam point): boiling point of pure water at standard pressure = 100°C = 212°F = 373 K. 1

Fixed points must be **reproducible** (always the same under the same conditions) and accessible anywhere in the world so that any two thermometers calibrated the same way will agree. 1

Calibration procedure: (1) Place the thermometer in a mixture of ice and water; mark the mercury level as 0°C (lower fixed point). (2) Place it in steam above boiling water; mark the mercury level as 100°C (upper fixed point). (3) Divide the gap between the two marks into 100 equal divisions. Each division = 1°C. 2

3(iv) — Liquid-in-glass vs. thermocouple; structural effects on performance 5 marks

Comparison table: 2

Aspect	Liquid-in-glass	Thermocouple
Thermometric property	Volume expansion of liquid	Change in emf (voltage)
Structure	Glass tube with bulb containing mercury or alcohol	Two different metal wires joined at a hot junction; cold junctions to voltmeter
Working principle	Liquid expands/contracts; column height indicates temperature	Voltage difference between hot and cold junctions proportional to temperature
Typical applications	Home, school, clinical (fever, lab)	Furnaces, kilns, engines, soil/water temperature in industry

Three ways structure of liquid-in-glass thermometer affects performance: 3

(1) **Capillary tube diameter (sensitivity):** Narrower bore means less liquid; same heat causes greater column rise, so sensitivity increases.

(2) **Bulb size (range):** Larger bulb holds more liquid; more expansion possible, so the thermometer covers a wider temperature range.

(3) **Type of glass (linearity):** Borosilicate glass has a more uniform, linear expansion coefficient than soda lime glass, so the scale divisions remain equally spaced (better linearity) across the full range.

OR

Five types of thermometers with thermometric properties and applications: 5

Thermometer type	Thermometric property	Preferred application
Laboratory/clinical (liquid-in-glass)	Volume expansion of liquid	Body temperature, lab work (simple and cheap)
Constant-pressure gas thermometer	Volume expansion of gas	Low-temperature research (wide range, linear)
Constant-volume gas thermometer	Pressure change of fixed gas mass	Standard laboratory calibration reference
Thermocouple	Change in emf between junctions	Industrial furnaces, kilns (very high temperatures)
Resistance thermometer (thermistor)	Change in electrical resistance	Precise scientific instruments, digital thermometers

Each row earns 1 mark for all three columns correct.